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User Interface Design

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CSE

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Chapter-4

User Interface Design

- Good User Interface design

What is User Interface Design?

- User interface is the front-end application view to which user interacts in order to use the software.
- User can manipulate and control the software as well as hardware by means of user interface.
- Today, user interface is found at almost every place where digital technology exists, right from computers, mobile phones, cars, music players, airplanes, ships etc.

- User interface is part of software and is designed such a way that it is expected to provide the user insight of the software.
- UI provides fundamental platform for human-computer interaction.
- UI can be graphical, text-based, audio-video based, depending upon the underlying hardware and software combination.
- UI can be hardware or software or a combination of both.

Types of User Interface

- 1.Command Line Interface:** The Command Line Interface provides a command prompt, where the user types the command and feeds it to the system. The user needs to remember the syntax of the command and its use.
- 2.Graphical User Interface:** Graphical User Interface provides a simple interactive interface to interact with the system. GUI can be a combination of both hardware and software. Using GUI, the user interprets the software.

When do the software become popular?

The software becomes more popular if its user interface is:

- Attractive
- Simple to use
- Responsive in short time
- Clear to understand
- Consistent on all interfacing screens

User Interface Design

- User Interface (UI) Design focuses on anticipating what users might need to do and ensuring that the interface has elements that are easy to access, understand, and use to facilitate those actions.
- UI brings together concepts from interaction design, visual design, and information architecture.

Characteristics of good user interface design

1. Simplicity :

1. User Interface design should be simple.
2. Less number of mouse clicks and keystrokes are required to accomplish this task.
3. It is important that new features only add if there is compelling need for them and they add significant values to the application.

2. Consistency :

1. The user interface should have a more consistency.
2. Consistency also prevents online designers information chaos, ambiguity and instability.
3. We should apply typeface, style and size convention in a consistent manner to all screen components that will add screen learning and improve screen readability. In this we can provide permanent objects as unchanging reference points around which the user can navigate.

3. Intuitiveness : (easy to learn)

1. The most important quality of good user interface design is intuitive.
2. Intuitive user interface design is one that is easy to learn so that user can pick it up quickly and easily.
3. Icons and labels should be concise and cogent. A clear unambiguous icon can help to make user interface intuitive and a good practice is make labels conform to the terminology that the application supports.

4. Graphical User Interface Design :

1. A graphic user interface design provides screen displays that create an operating environment for the user and form an explicit visual and functional context for user's actions.
2. It includes standard objects like buttons, icons, text, field, windows, images, pull-down and pop-up screen menus.

5. Aesthetic and attractive :

As we all know attractive things gain more attention. Thus, a good user interface should be attractive to use. Thus, graphics-based user interfaces are in great demand over text-based interfaces.

6. Feedback : (how fast you are getting the response)

Providing remarks to the moves of the person facilitates person to apprehend processing of the system. If any request of user takes more than a few seconds then user starts to panic, that is what is happening, if the proper feedback is providing to user, then he must know about his actions. Thus, a good user interface must contain feedback about the processing.

7. Error recovery (undo)

Error could be very common, all people can dedicate an blunders even specialists also can dedicate mistakes. Therefore, it's also a responsibility of a great person interface to offer a undo facility in order that person can get better their errors at the same time as use of the interface. If the mistakes can't be recovered through users, they experience irritated, helpless, and low.

Best Practices for Designing an Interface

Everything stems from knowing your users, including understanding their goals, skills, preferences, and tendencies. Once you know about your user, make sure to consider the following when designing your interface:

- **Keep the interface simple.** The best interfaces are almost invisible to the user. They avoid unnecessary elements and are clear in the language they use on labels and in messaging.
- **Create consistency and use common UI elements.** By using common elements in your UI, users feel more comfortable and are able to get things done more quickly. It is also important to create patterns in language, layout and design throughout the site to help facilitate efficiency. Once a user learns how to do something, they should be able to transfer that skill to other parts of the site.

Best Practices for Designing an Interface

- **Strategically use color and texture.** You can direct attention toward or redirect attention away from items using color, light, contrast, and texture to your advantage.
- **Use typography to create hierarchy and clarity.** Carefully consider how you use typeface. Different sizes, fonts, and arrangement of the text to help increase scan ability and readability.
- **Be purposeful in page layout.** Consider the spatial relationships between items on the page and structure the page based on importance. Careful placement of items can help draw attention to the most important pieces of information and can aid scanning and readability.

Best Practices for Designing an Interface

- **Make sure that the system communicates what's happening.** Always inform your users of location, actions, changes in state, or errors. The use of various UI elements to communicate status and, if necessary, next steps can reduce frustration for your user.
- **Think about the defaults.** By carefully thinking about and anticipating the goals people bring to your site, you can create defaults that reduce the burden on the user. This becomes particularly important when it comes to form design where you might have an opportunity to have some fields pre-chosen or filled out.

References

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THANK YOU